

Bayesian Methods

Significant Strikes Markets

UFC-MMA Betting · Originator Pricing

Date : 2025 that's all I remember

Introduction

During a Saturday shift at the trading room, I saw some bets coming through the tickers that intrigued me. I had to open them and check whether the customers were fine, or whether they were taking advantage of odds we had no business offering..... It was the second option.

The customer was on UFC player props. He had picked a fighter — I cannot remember the name — to land Over X.5 significant strikes. The stakes were maxed. My first thought was: who puts that much money on a market this volatile?

Significant strikes markets are usually up only two days before the fight. Sportsbooks do not like holding them longer. The lines move, the variance is enormous, and the precautionary measures are familiar: high juice, and at the softer shops, lower limits than you would expect for a top-tier sport.

Looking closer, we were offering an 11% arbitrage against another book. Main lines across shops were not close to each other. That was the moment I decided to build a simple model of my own — not to copy a sharp, and not to shade a line I did not believe. I wanted to originate a price, then see which sportsbooks were closest to it and which ones were giving the game away.

Originator Pricing, Not Top-Down

Most of what a sportsbook does with these props is top-down. Somebody posts a number. The rest of the market copies it, trims it, and adds juice. If you start from that number and “adjust,” you are not pricing. You are inheriting someone else’s error and dressing it up.

Originator pricing goes the other way. The inputs are the fighter’s significant-strike history, the minutes those strikes were landed in, a prior on the strike rate, and a duration for the fight you are pricing. The output is a predictive distribution: how many significant strikes we should expect, and the probability of every Over/Under hanging on the board.

The sportsbook line is not an input. It is the thing we compare against at the end. If bet365 hangs 53.5 and the model says the mass sits around 30 to 40, we do not walk our number toward 53.5 because “that is where the market is.” We take the under, or we pass. That is the whole point of building the model.

A point estimate is not enough for this. We need a full distribution. That is why the Gamma-Poisson setup is the right language: it gives a posterior on the strike rate and, after a duration, a Poisson predictive for the count.

Why Bayesian Methods?

In my final academic year I took a decision-theory class and met conjugate priors properly for the first time. Being able to put a belief into a model is powerful and dangerous. In sports betting, public narratives and personal bias are the default, not the exception. A prior can bake that in and ruin you, or it can encode a real stylistic fact that the market is too slow to price.

Khabib Nurmagomedov is the clean example. He forced opponents into wrestling and grappling and crushed their striking output. If you can measure how much the opponent's strikes per minute drop in that kind of matchup, you can put that into the prior the next time a comparable wrestler takes a striker down for fifteen minutes.

I still run every fight with a flat prior next to the informed one. If the posterior barely moves, the data were doing the work. If it moves a lot, I want to see exactly how much my story is worth. The section on priors later in this note is about that knob: α and β , mean versus conviction.

I use two models. Same Poisson idea. Different way of turning a career into a rate.

Model 1 — Gamma-Poisson with Arithmetic Mean

We assume significant strikes follow a Poisson process:

$$X \sim \text{Poisson}(\lambda \cdot t)$$

λ is the average number of significant strikes per minute. t is the expected duration of the upcoming fight. From each historical bout:

$$STR_PM = \text{Significant Strikes} / \text{Duration}$$

The arithmetic model treats every fight as one observation of that rate, whether it lasted six minutes or fifteen:

$$\lambda \mid \text{data} \sim \text{Gamma}(\alpha + \Sigma STR_PM, \beta + n)$$

n is the number of fights. A one-round KO and a decision go into the posterior with the same weight. That is the feature and the bug. Chaotic short fights can dominate the rate — which is sometimes how the market itself behaves.

Model 2 — Gamma-Poisson with Weighted Mean

The same Poisson assumption, taken seriously. Fight i contributes strikes over T_i minutes:

$$X_i \sim \text{Poisson}(\lambda \cdot T_i)$$

Pool the counts and the clock:

$$\lambda \mid \text{data} \sim \text{Gamma}(\alpha + \Sigma \text{strikes}, \beta + \Sigma \text{minutes})$$

A fifteen-minute fight is worth more than a six-minute fight, which is what the Poisson process says. This is a statistically coherent update. I still run the arithmetic version beside it. The market

sometimes overweights the same highlight-reel short fights the arithmetic model does, and I want both numbers in front of me before I originate a price.

Worked Example: Talbott vs Lima

Payton Talbott vs Felipe Lima. Flat prior $\lambda \sim \text{Gamma}(1, 1)$ on both fighters. No matchup story yet — just the data and duration.

Talbott

Fight	Sig. Strikes	Duration (min)	Strikes / min
vs Barcelos	49	15	3.27
vs Ghemmouri	13	6	2.17
vs Saaiman	79	6	13.17
vs Aguirre	28	11	2.55
vs Cortez Jr	145	15	9.67

Lima

Fight	Sig. Strikes	Duration (min)	Strikes / min
vs Johns	55	15	3.67
vs Naimov	37	12	3.08

Arithmetic update. Talbott: $\Sigma \text{STR_PM} = 30.81$, $n = 5$, so $\lambda \mid \text{Talbott} \sim \text{Gamma}(31.81, 6)$, posterior mean 5.30 strikes per minute. Lima: $\Sigma \text{STR_PM} = 6.75$, $n = 2$, so $\lambda \mid \text{Lima} \sim \text{Gamma}(7.75, 3)$, posterior mean 2.58.

Weighted update. Talbott: 314 strikes in 53 minutes, $\lambda \mid \text{Talbott} \sim \text{Gamma}(315, 54)$, posterior mean 5.83. Lima: 92 strikes in 27 minutes, $\lambda \mid \text{Lima} \sim \text{Gamma}(93, 28)$, posterior mean 3.32.

Duration is the remaining piece. We do not invent minutes. We take the Over/Under 2.5 the book is offering — here bet365, over 1.40 and Under 2.85 — remove the juice, split the under probability equally between a round-1 finish and a round-2 finish, give the over to round 3, and convert expected rounds into minutes. That is about 12.5 minutes. It is a crude clock. It is also the clock implied by the same sportsbook we are trying to beat, which is the honest duration to use if we are against that board.

Then we draw λ from each posterior 10,000 times and simulate

$$\text{Significant Strikes} \sim \text{Poisson}(\lambda \cdot 12.5)$$

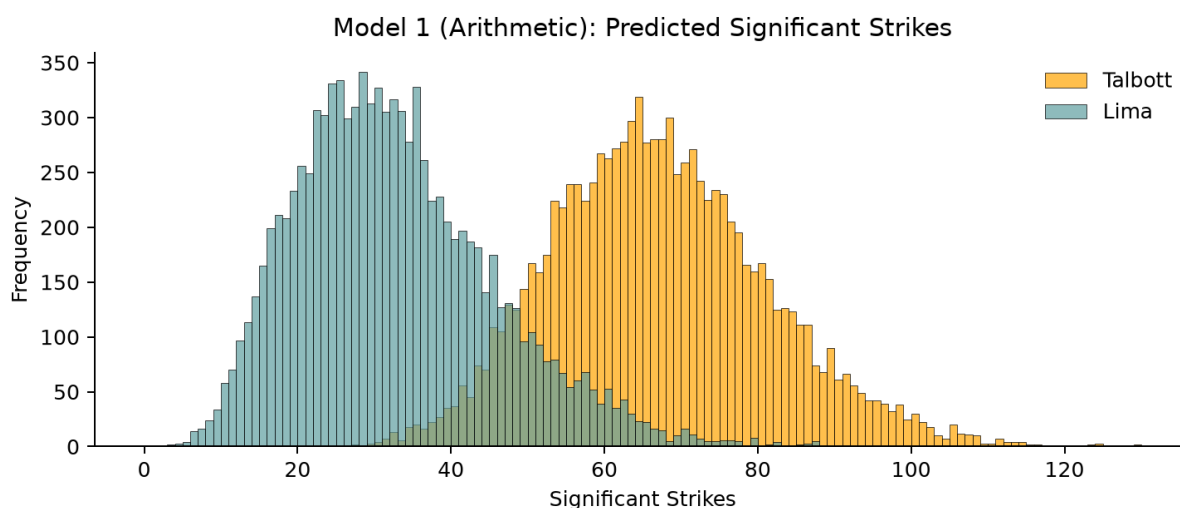


Figure 1. Model 1 (arithmetic). Fighter 1 = Talbott, Fighter 2 = Lima.

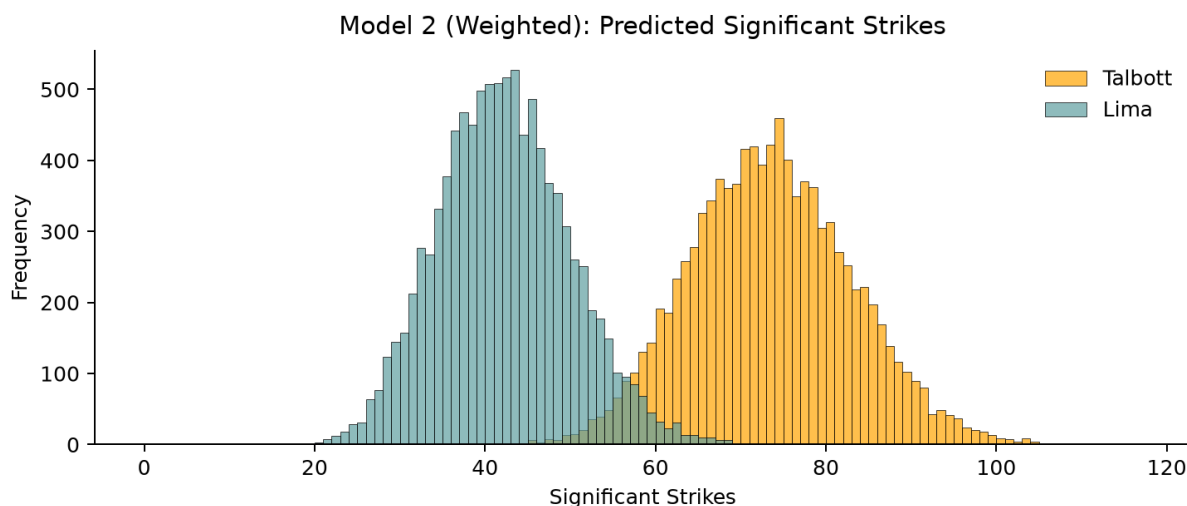


Figure 2. Model 2 (weighted / exposure). Fighter 1 = Talbott, Fighter 2 = Lima.

Arithmetic medians: Talbott 66, Lima 31. Weighted medians: Talbott 73, Lima 41. Bet365 hung Lima's main line at 53.5. Both models sit well under it. The book overestimated his striking output, so the originated price is the under. Naturally we went in strong — a massive €2. Skin in the game, as Taleb would have it.

$P(\text{Lima} > 53.5)$ is about 6–7% in both models. That is not a close line. That is a misprice. The two models do not agree on how low. Arithmetic is harsher on Lima because two fights and a Gamma (1, 1) prior already shrink him toward one strike per minute. Weighted respects the 27 minutes of actual striking and lands near 41. When they disagree, I trust the weighted number more as a rate, and I still want the arithmetic number in front of me as a stress test.

Priors: Mean Versus Conviction

Everything above used Gamma (1, 1). That is the “I do not want to say much” prior. The moment you stop using it, you have to know what you are turning.

In the shape-rate Gamma this model uses:

$$E[\lambda] = \alpha / \beta \quad \text{Var}(\lambda) = \alpha / \beta^2$$

The ratio is the mean. It is not the variance. Gamma (1, 2) and Gamma (100, 200) both say “I expect 0.5 strikes per minute.” They do not say it with the same mouth.

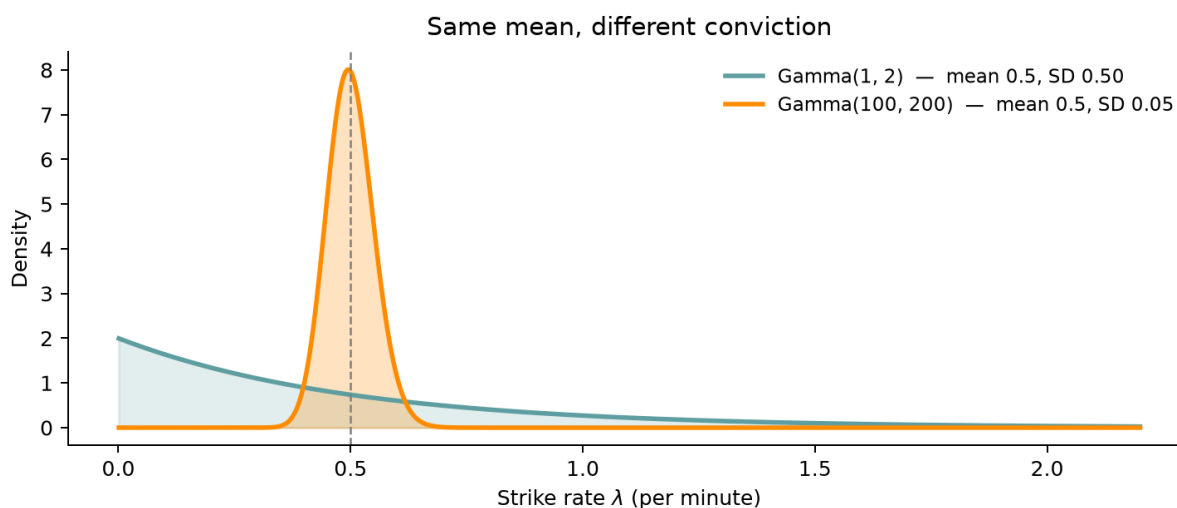


Figure 3. Same mean, different conviction. The prior’s force lives in how large α and β are together, not in the ratio α/β .

Gamma (1, 2): mean 0.5, standard deviation 0.50. A shrug. Gamma (100, 200): mean 0.5, standard deviation 0.05. A hill you die on.

α and β are fake history you mix in before the real fights. In the weighted model, α is a prior strike count and β is prior minutes. In the arithmetic model, α is a prior sum of STR_PM and β is a prior number of fights. The posterior mean is a weighted average of what you believed and what the data did:

$$E[\lambda \mid \text{data}] = [\beta / (\beta + T)] \cdot (\alpha / \beta) + [T / (\beta + T)] \cdot (S / T)$$

T is observed exposure — minutes in the weighted model, fights in the arithmetic model. β is the conviction knob. only aims for that conviction. Lima has two fights and 27 minutes. He is the fighter the prior can bully.

Prior	Mean	SD	Prior weight	Posterior
-------	------	----	--------------	-----------

Gamma (1, 1)	1.00	1.00	4%	3.32
Gamma (1, 2)	0.50	0.50	7%	3.21
Gamma(100, 200)	0.50	0.05	88%	0.85

Lima, weighted model: 92 strikes in 27 minutes, MLE = 3.41 per minute. Prior weight is $\beta / (\beta + T)$. Same mean on the last two rows; only the strong prior actually uses it.

Gamma (1, 2) barely nudges him. Gamma (100, 200) steamrolls him. You are no longer estimating Lima; you are repeating the prior and sprinkling a little data on top. That is the difference between changing the mean and changing the level of conviction.

Two more things worth being honest about.

First, Gamma (1, 1) is not uninformative. It is Exponential (1). Mean of one strike per minute — five per round, fifteen per fifteen-minute fight. With two fights, arithmetic Lima is already one-third prior. The “flat” arithmetic posterior mean is 2.58, not the 3.38 the two fights actually averaged. If you want the data to speak, use something like Gamma(0.01, 0.01), or keep the mean near a typical UFC pace and keep β tiny: Gamma(4, 1) or Gamma(5, 1).

Second, the same (α, β) is not the same prior in the two models. $\beta = 2$ in the weighted model is two fake minutes. $\beta = 2$ in the arithmetic model is two fake fights. Lima only has two real fights, so that prior is already half the posterior. This is one reason the arithmetic Lima number sits lower than the weighted one even before you get clever with matchup priors.

If you have a real stylistic view, set it like a trader, not like a mystic:

1. Pick the rate you believe — the mean α/β .
2. Pick how many minutes or fights that view is worth — β .
3. Set $\alpha = \text{mean} \times \beta$.

Khabib-type wrestler: you think the opponent’s output is 2.0 per minute, and that view is worth about one fake fight. Then $\beta = 15$, $\alpha = 30$. Mean 2.0, standard deviation 0.37. An opinion. If you set $\beta = 200$ and $\alpha = 400$, you have decided the market and the data can no longer talk. Run that prior next to Gamma(1, 1). The gap is the price of your story. The next section is that recipe used on a real fighter.

A Matchup Prior: Khamzat Chimaev

Take a Khamzat Chimaev fight. He is known to wrestle. The story is not that he is a low-output striker. The story is that the other man does not get to hit. So, this prior is not on Khamzat's λ . It is on the opponent's λ : significant strikes the opponent will land per minute while Khamzat is wrestling him.

In this model those two numbers have a sentence:

α = fake significant strikes already seen on Khamzat while being wrestled

β = fake minutes of that wrestling already seen

So Gamma (α , β) means: I have already watched β minutes of Khamzat sitting on someone, and they landed α significant strikes. The mean α/β is the smothered rate. β is how hard I insist this fight is that fight.

When he actually wrestles from the start, the first round is not a 4.0 SLpM striker having an off night. It is a man on his back.

Fight	Opp. sig. strikes	Time	Rate / min
vs Whittaker	2	3:34	0.56
vs Holland	0	2:13	0.00
vs Usman (R1)	3	5:00	0.60

Whittaker and Usman are high-volume strikers by career. Against a wrestling Khamzat in round one they land about 0.5 significant strikes per minute — two or three in a full round. Usman R1 is about 4:35 of control.

Burns is the other tape: Khamzat did not smother him, and Burns landed 119 in fifteen minutes. If you think the fight is Burns, do not use this prior. If you think it is Whittaker, Holland, Usman round one, or the Dricus wrestling clinic, you do.

The first-round default, if Khamzat shoots early, is a mean of 0.5 per minute — 2.5 significant strikes in a five-minute round — and a conviction of one Khamzat first round, $\beta = 5$. Then $\alpha = 0.5 \times 5 = 2.5$:

$\lambda \text{ opponent} \sim \text{Gamma}(2.5, 5)$

In English: I have already seen 2.5 significant strikes in five minutes of Khamzat wrestling. This round will look like that. That is an opinion, not a lock. The standard deviation is $\sqrt{2.5 / 5} \approx 0.32$, so most of the prior mass sits around 0.2 to 1.1 per minute — about 1 to 5.5 strikes if the round lasts five minutes. Same mean, harder conviction, is just a larger β :

Story	β (fake min)	$\alpha = 0.5 \times \beta$	What you are saying
Shrug	5	2.5	Worth one Khamzat R1
I have watched this movie	10	5	Worth two such rounds
This is his whole game	15	7.5	A full 15-min smother

The trap is feeding the opponent's entire stand-up career into the likelihood and only adding five fake minutes of Khamzat. Posterior mean is still

$[\beta / (\beta + T)] \cdot 0.5 + [T / (\beta + T)] \cdot (\text{career rate})$

A 4.0 SLpM striker with 60 minutes of career data and $\text{Gamma}(2.5, 5)$ gives the prior only 8% of the posterior. The number lands around 3.7. You said “Khamzat wrestles.” The posterior barely heard you. Career minutes ate the story.

To actually originate the under, the likelihood has to be first-round data, not fifteen-minute kickboxing fights. Then T is 20 or 30 minutes of round one, not 80 minutes of career, and $\text{Gamma}(2.5, 5)$ has a real pull. That is also the market you are pricing. If you insist on using career totals, β has to be huge: dragging a 4.0 guy to about 1.5 with a 0.5 mean needs roughly 70% prior weight, so $\beta \approx 150$ and $\alpha \approx 75$ against $T = 60$. The sentence is then: Khamzat-wrestling minutes are worth two and a half of this man’s careers. You can say that. You must know you said it.

The cleaner originator version does not start from this opponent’s career at all. The relevant sample is what people land on Khamzat when he wrestles: Whittaker 2 in 3:34, Holland 0 in 2:13, Usman 3 in 5. That is the likelihood for output against wrestling Khamzat. This opponent only bumps the mean if he is a genuine scrambler. You are not shading a 19.5 that was built from career SLpM times five. You never started there.

What goes into the model, opponent, first round, Khamzat wrestles early: shape = 2.5, rate = 5. Run $\text{Gamma}(1, 1)$ on the same first-round data next to it. If the book is hanging 18.5 off career pace and the informed posterior sits around 3 to 6, that is the under. If both priors still say 16, Khamzat is not wrestling this one, or the data were the wrong minutes.

Do not put this prior on Khamzat’s own strikes. His λ is a different story — ground and pound versus stall. This prior is only: the other guy does not get to hit.

Note — 1st Round Significant Strikes

The full-fight model has a duration problem. We manufacture duration from Over/Under 2.5. That mapping is crude: it does not know about early round-3 stoppages, decisions versus knockouts, or five-round main events. It is a necessary evil if we want a total-fight number.

First-round markets do not need it. We originate a five-minute round. Historical first rounds that ended early are entered with their actual minutes — the same way the full-fight model uses actual duration — and the prediction is for a full five-minute first round. That is a cleaner Poisson problem. The clock is not the debate.

It is also a better trading problem, and this is the point. There is a class of fighters who take opponents down early. Their first-round significant strikes are not a scaled version of their career rate. They are much lower. The opponent’s first-round significant strikes are lower too: you cannot hit what you are holding on the fence.

Books still hang first-round lines that look like career SLpM times five minutes, or like a generic slice of the full-fight total. A wrestler who averages 4.0 significant strikes per minute over a career gets a first-round offer around 19.5 or 20.5. If his actual first-round history is 8, 11, 9, 12 because the game plan is shoot, clinch, and empty the gas tank later, the originator number from

first-round data sits well under the offer. The same holds for the striker who must play on his back for four minutes. That is not a small edge. That is the market using the wrong exposure.

This is originator pricing in the place it is easiest to do. You do not have to win an argument about whether the fight lasts 9 minutes or 14. You have to be right about round one. The inputs are first-round significant strikes, first-round durations, and — when the takedown story is real — a prior that is allowed to know it. The two aggregation methods stay the same. Weighted still respects minutes. Arithmetic still lets a ninety-second explosion count as a full observation. I want both, especially in round one, because the short finishes are exactly where the arithmetic model and the sportsbook can both get drunk.

Khamzat is the template. I do not wait for the posterior to discover the wrestle from three mixed career fights. I put $\text{Gamma}(2.5, 5)$ on the opponent's first-round rate, I run $\text{Gamma}(1, 1)$ next to it, and I originate the first-round under off first-round data. That is the trade. The full-fight total can wait.

Skin in the Game

The €2 on Lima was a joke. These were not. Same story, real stakes: Khamzat wrestles, the other man does not strike, originate the first-round under.

Fight	Market	Odds	Stake	Return
DDP vs Khamzat	DDP U17.5 R1	1.83	€100	€183.34
Strickland vs Khamzat	Strickland U9.5 R1	1.83	€150	€275.00
Strickland vs Khamzat	Strickland U10.5 R1	1.83	€50	€91.67

All three first-round significant-strikes unders. Combined stake €300. Combined return €550.01. All won.

Dricus du Plessis, 17 August 2025. The book hung 17.5 significant strikes in round one on a man Khamzat was about to hold on the mat. Career SLpM times five is how you get to a number like that. The originator number, once you admit he is being wrestled, is nowhere near 17. Under 17.5 at 1.83. €100. Won.

Dricus Du Plessis Under 17.5	1.83
Dricus Du Plessis vs Khamzat Chimaev 17/08/2025	
A Significant Strikes Landed in Round 1	Won
Single 1 x €100.00	

Total Stake €100.00
Return €183.34

Figure 4. Dricus du Plessis Under 17.5 significant strikes, round 1. Khamzat Chimaev, 17/08/2025. Settled: Won.

Sean Strickland, 8 May 2026, the day before UFC 328. The offer collapsed all the way to 6.5 from 10.5. That is the market finally seeing the wrestle and still leaving a number a striker of Strickland's reputation should not be under if the fight stays standing. It was not staying standing. €150 on Under 9.5, €50 on Under 10.5, both 1.83. Both won.

08/05/2026 15:13:20	
Sean Strickland Under 9.5	1.83
€150.00 Single	
Stake €150.00	Return €275.00
Bet Details >	

08/05/2026 15:10:29	
Sean Strickland Under 10.5	1.83
€50.00 Single	
Stake €50.00	Return €91.67
Bet Details >	

Figure 5. Sean Strickland Under 9.5 and Under 10.5 significant strikes, round 1. 08/05/2026, Khamzat Chimaev. Both settled as wins.

This is what originator pricing looks like when the prior is a real matchup and the market is still hanging a career rate. You do not need a miracle. You need Khamzat to wrestle, and a book still pricing round one as if he will not.

The Hard Part: Finding the Fight in Time

The model is the easy bit. The difficulty with this betting strategy is operational. Significant strikes markets go up late — usually two days before the event, sometimes later — and they are not posted on every bout. In practice they live on the main card. You are not scanning a twelve-fight slate for mispriced λ . You are waiting for a handful of lines, and you have to find the offering in time.

Even then you need the matchup to have the characteristic. Most main cards do not. You need to be lucky enough that the fight in front of you is one of the two movies this model is built for, and honest enough to pass when it is not.

The first movie is the wrestle. Khamzat, Khabib, Asu Almagbayev anyone who takes the fight down early. Opponent first-round significant strikes come in well under the career rate. That is the under.

The second is the same but with two strikers who will go toe to toe and keep standing but stay defensive. Michael “Venom” Page is the clean example — point fighting, long range, the fight stays at distance. There is no hidden grappling tax the constant distance control by MVP.

Using the Models

We originate. We do not shade. Weighted is the coherent rate. Arithmetic is the equal-fight stress test. Gamma(1, 1) is the baseline. An informed prior is a matchup view with an explicit β — for Khamzat, 2.5 strikes in 5 minutes of wrestling on the opponent, not a vibe. Full-fight lines need a duration. First-round lines do not, and they are where early-takedown fighters leave their opponents systematically overpriced. The offering is scarce: main card, late, and only when the matchup actually has the characteristic. Read it first. Bet only if it is true.

Attack the board with the predictive distribution — Over 15.5, 20.5, 25.5, most significant strikes — not with a single median. If both models and both priors still say the same side, that is as much conviction as this market deserves.